

SOMERSET, NJ, USA

WMO: 722247

Lat: 40.624N

Lon: 74.669W

Elev: 32

StdP: 100.94

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54785

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-10.9	-21.4	0.6	-10.8	-18.8	0.7	-8.2	9.2	-1.8	8.3	-1.7	1.4	310	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	33.1	23.7	31.8	23.0	30.4	22.2	25.2	30.6	24.4	29.2	23.7	28.0	2.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.7	27.5	22.9	17.7	26.4	22.4	17.1	25.9	77.2	30.6	73.8	29.1	70.8	28.1	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.7	5.5	DB	-16.8	35.5	3.7	1.9	-19.5	36.8	-21.7	37.9	-23.8	39.0	-26.5	40.3	(4)
(5)			WB	-17.4	26.8	3.5	0.9	-19.9	27.5	-22.0	28.0	-23.9	28.6	-26.4	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-1.2	0.0	4.6	10.9	16.2	20.9	23.8	22.9	19.2	12.5	6.9	2.0	(6)
(7)		DBStd	9.68	5.42	4.98	4.89	4.64	4.17	3.32	2.63	2.65	3.53	4.63	4.60	4.79	(7)
(8)		HDD10.0	1248	348	282	179	42	3	0	0	0	27	116	252	(8)	
(9)		HDD18.3	2952	606	514	427	227	93	14	1	31	189	344	506	(9)	
(10)		CDD10.0	1841	1	2	11	69	196	328	427	399	276	104	23	4	(10)
(11)		CDD18.3	502	0	0	0	4	28	92	169	143	57	8	0	0	(11)
(12)		CDH23.3	4831	0	1	5	105	342	918	1634	1255	509	61	2	0	(12)
(13)		CDH26.7	1725	0	0	0	29	104	314	665	448	154	11	0	0	(13)
(14)	Wind	WSAvg	1.8	2.1	2.3	2.5	2.4	1.7	1.4	1.3	1.1	1.4	1.6	1.9	2.0	(14)
(15)	Precipitation	PrecAvg	1277	92	74	114	99	99	121	124	114	119	117	89	116	(15)
(16)		PrecMax	1655	195	164	254	275	229	243	246	233	312	345	168	212	(16)
(17)		PrecMin	1018	43	24	23	39	20	29	28	36	28	17	25	29	(17)
(18)		PrecStd	155	38	34	49	50	43	55	55	51	65	80	37	45	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.8	23.5	30.3	31.9	33.8	35.5	34.5	33.2	27.9	22.3	19.0	(19)
(20)			MCWB	13.4	13.1	15.2	18.2	21.8	23.7	24.1	24.5	23.2	20.5	16.6	15.7	(20)
(21)		2%	DB	12.8	14.0	19.1	26.3	29.6	31.9	33.5	32.4	30.5	24.9	19.8	15.0	(21)
(22)			MCWB	10.2	9.0	12.4	16.0	20.8	23.0	23.8	23.7	21.5	18.8	15.5	12.7	(22)
(23)		5%	DB	9.1	10.4	15.8	23.0	27.3	30.0	32.0	30.8	28.1	22.6	17.4	12.0	(23)
(24)			MCWB	6.1	6.3	9.9	14.7	19.1	21.8	23.3	22.6	20.4	17.8	13.4	9.7	(24)
(25)		10%	DB	6.7	7.3	12.7	20.0	24.7	28.4	30.4	29.2	26.3	20.6	15.1	9.6	(25)
(26)			MCWB	4.2	3.6	8.2	12.9	17.8	20.9	22.5	21.7	19.9	16.4	11.6	6.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	15.1	16.2	20.0	23.3	25.1	26.5	26.6	24.9	22.4	18.4	16.6	(27)
(28)			MCD B	16.2	18.4	22.0	26.6	29.4	31.3	32.8	32.1	30.4	25.2	21.0	18.3	(28)
(29)		2%	WB	10.2	10.2	13.6	18.0	21.9	24.1	25.4	25.2	23.6	20.8	16.5	13.0	(29)
(30)			MCD B	12.4	14.0	17.8	24.3	27.5	29.6	31.0	29.8	27.3	23.7	18.9	14.8	(30)
(31)		5%	WB	6.8	6.5	11.0	16.0	20.6	23.3	24.6	24.1	22.5	19.0	14.4	10.3	(31)
(32)			MCD B	8.6	9.3	14.6	21.4	25.2	28.2	29.4	28.3	25.7	21.7	16.8	11.9	(32)
(33)		10%	WB	4.4	4.3	8.6	14.1	19.2	22.2	23.8	23.3	21.5	17.2	12.3	7.3	(33)
(34)			MCD B	6.6	6.7	12.3	18.6	23.3	26.6	28.1	26.9	24.6	19.4	14.4	8.9	(34)
(35)	Mean Daily Temperature Range	MDBR		9.7	10.7	11.1	12.9	12.5	12.2	12.0	11.7	12.3	12.1	11.5	9.2	(35)
(36)		5% DB	MCD BR	13.2	15.7	16.4	18.3	16.1	14.3	13.8	13.4	14.8	14.4	14.3	12.3	(36)
(37)			MCWBR	10.7	11.3	10.6	9.4	8.0	6.7	5.6	5.6	7.0	8.5	9.8	10.4	(37)
(38)		5% WB	MCD BR	12.0	14.6	14.2	15.4	13.2	12.6	11.4	10.7	11.2	11.3	11.6	11.0	(38)
(39)			MCWBR	10.4	11.1	10.7	9.1	7.2	6.5	5.3	5.2	6.3	7.7	9.4	10.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.309	0.325	0.360	0.384	0.430	0.492	0.489	0.480	0.395	0.362	0.328	0.308	(40)
(41)		taud		2.452	2.400	2.373	2.339	2.240	2.105	2.127	2.143	2.352	2.455	2.506	2.515	(41)
(42)		Ebn at Noon		851	888	890	888	851	796	794	789	842	835	823	826	(42)
(43)		Edh at Noon		78	95	110	121	137	158	153	146	111	90	75	69	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.86	2.77	3.67	4.74	5.25	5.76	5.91	5.05	4.25	2.97	2.10	1.58	(44)
(45)		RadStd		0.16	0.29	0.30	0.46	0.57	0.51	0.29	0.33	0.47	0.31	0.19	0.16	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	+68

Nomenclature: See separate page